

Education

Bachelor of Science in Data Science Iowa State University, Ames, IA	Aug 2023 – Present
Master's of Science in Mechanical Engineering Iowa State University, Ames, IA	Jan 2025 – Present

EXPERIENCE

Undergraduate Research Assistant <i>Translational AI Center, Iowa State University</i> • Collaborating with Professor Soumik Sarkar on leveraging LLMs in finance to enhance decision-making processes. • Gaining insights into AI integration in financial applications, and developing advanced skills in machine learning.	Apr 2024 – Present
Undergraduate Software Developer <i>Iowa State University Department of Agricultural and Biosystems Engineering</i> • Worked with Professor Ryan McGhee on developing a versatile chatbot frontend to improve UI and experience. • Engaged in software development and user interface design.	June 2024 – December 2024
Undergraduate Research Volunteer <i>ISU Department of Computer Science</i> • Explored the behavior of large language models with knowledge conflicts under Professor Qili's guidance. • Developed key skills in data research and analysis.	Apr 2024 – May 2024

PUBLICATIONS & PRESENTATIONS

SAGE: Scalable Agentic Grounded Evaluation for Crop Disease Diagnosis

AI Scientist Summer Workshop, Microsoft Research New England, Cambridge, MA, Poster

- Selected for poster presentation at the AI Scientist Summer Workshop hosted by Microsoft Research New England, August 4, 2026.
- Co-first-authored an autonomous, symptom-grounded vision-language agent for crop-disease diagnosis, evaluated on the largest plant-disease image-symptom dataset to date; stepwise diagnostic reasoning improved accuracy by +16.2 points and generalizes to unseen crops without retraining.

Intelligent Health Intervention System for Syndemic Management Among University Students: An AI-Driven Approach Using Reinforcement Learning

AAAI 2026 Spring Symposium Series - "AI in Business: Intelligent Transformation and Management," Oral Presentation

- Accepted for oral presentation at AAAI 2026 Spring Symposium, April 7-9, 2026, San Francisco, CA.
- Developed AI-powered chatbot using reinforcement learning for personalized health interventions addressing syndemic conditions in university students.
- Implemented Proximal Policy Optimization algorithms integrated with fine-tuned LLMs and medical knowledge databases for managing communicable diseases, non-communicable diseases, and mental health.

An Agentic Framework for Therapeutic Reasoning in Oncology

NE Agents Day 2026 Workshop, Poster

- Developed an agentic reasoning framework specifically for oncological therapeutic decision-making.
- Integrated knowledge graphs and multi-agent systems to support clinical reasoning processes.

An Agentic Framework for Mechanistic Therapeutic Reasoning in Oncology via Pathway-Grounded Tree-of-Thought Inference

ACM BCB 2026 Posters (26th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics)

- Presented advanced agentic framework combining pathway analysis with tree-of-thought reasoning for oncology.
- Integrated mechanistic biological pathways with LLM-based reasoning for therapeutic inference.

Federated In-Context Prompt Selection for Multi-Modal 3D Dental Imaging: A Theoretical Framework with Privacy-Preserving Guarantees

ODIN 2025 (Oral and Dental Image aNalysis) Workshop @ MICCAI 2025 (29th International Conference on Medical Image Computing and Computer Assisted Intervention), Poster

- Developed a novel federated learning framework for multi-modal 2D and 3D dental imaging with privacy-preserving guarantees.
- Key contributions: cross-modal prompt alignment, hierarchical multi-modal optimization, and Byzantine-resilient aggregation with differential privacy.

LLM-based Insights into How Market Orientation and Brand Storytelling Affect Investors' Evaluation of Startup Pitches

REU Summer Symposium 2025, Poster; TrAC Day 2025, Poster & Flash Talk

- Presented research poster at REU Summer Symposium 2025 while serving as a peer mentor to Taj Singh, enhancing both academic and personal growth within the Translational AI Center.
- Delivered a flash talk at TrAC Day 2025, exploring how passion, responsiveness, and storytelling drive investor decisions using GPT-4o and Shark Tank data.

Enhancing Medical AI Safety Through Multi-Agent Evaluation and Iterative Refinement

The 16th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB) 2025, Pennsylvania, USA, Poster

- Developed a novel multi-agent refinement framework to enhance the ethical, safety, and regulatory compliance of medical LLM outputs.
- Framework pairs DeepSeek R1 and Med-PaLM as generative models with LLaMA 3.1 and Phi-4 as evaluator agents, iteratively refining responses against the American Medical Association's nine Principles (AMA-9) and a five-tier Safety Risk Assessment (SRA-5).
- Strengthened reliability and alignment of medical LLM outputs, addressing real-world deployment challenges.

LlamaHealth: Assessing LLaMA's Performance in Multi-Class Healthcare Test Result Prediction

ICERIE 2025, Shahjalal University of Science and Technology, Sylhet, Bangladesh, Poster

- Earned Certificate of Participation; reinforced focus on responsible AI for healthcare.
- Investigated LLaMA's multi-class prediction capabilities in healthcare datasets.

Benchmarking Bioemu-1: Establishing a Baseline for Protein Structural Ensemble Prediction

AICHe Annual Meeting 2025, Boston, USA, Poster

- Presented in the Computational Molecular Science and Engineering Forum poster session.

Leveraging Vision Language Models for Specialized Agricultural Tasks

Proceedings of the 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

- Explored how vision-language models improve performance on specialized agricultural benchmarks.

The Impact of Market Orientation and Brand Storytelling on Shark Tank Evaluations

Journal of Business & Industrial Marketing (JBIM)

- Analyzed how storytelling and market orientation influence investor evaluation using LLM analysis.

INTERNSHIPS

Undergraduate Summer Research Intern – Translational AI Center

May 2024 - July 2024

HONORS & AWARDS

Dean's High Impact Award for Undergraduate Research (Fall 2025)

- Recognized for outstanding contributions to undergraduate research at Iowa State University.
- Highlighted for innovation and rigor in research area/project.
- Mentored or collaborated with peers to enhance research outcomes.

Dean's List, College of Liberal Arts and Sciences (Fall 2024) - *High Impact Academic Recognition*

- Achieved distinction for academic excellence across all courses at Iowa State University.
- Maintained high GPA while actively engaging in research and extracurricular activities.

Best-In-Show, George Washington Carver Poster Challenge (2025)

- Awarded top recognition for research poster presentation at the annual Carver Poster Challenge.
- Selected from participants based on clarity, impact, and originality of research.

Borlaug Poster Competition (2025)

- Developed research under TrAC – Translational AI Center exploring how AI and FinTech empower entrepreneurs addressing global food and sustainability challenges.
- Recognized as Undergraduate division 1st place by Iowa State University College of Agriculture and Life Sciences for connecting technology, entrepreneurship, and global food systems.

Bangladesh–Sweden Trust Fund Travel Grant (2025)

- Competitive international grant supporting research travel and presentation opportunities.
- Enabled participation in conference/seminar/workshop and networking with global researchers.

ISU LAS Travel Grant (\$400, AIChE 2025)

- Awarded by the College of Liberal Arts and Sciences, Iowa State University.
- Supported conference presentation at AIChE 2025 and engagement with leading experts.

ACADEMIC SERVICE

Reviewer, CV4Animals 2026 (CVPR 2026 Workshop)

- Reviewed computer vision applications for animal identification and monitoring in livestock systems.
- Evaluated non-invasive identification technologies for group-housed animals.

Reviewer, AI4GOOD Workshop 2026

- Reviewed submissions on AI applications for social good and global development challenges.
- Contributed to advancing responsible and impactful AI research.

Reviewer, AAMAS 2026 Workshop AI4CNI (Autonomous Agents and Multiagent Systems)

- Reviewed papers on agent-based systems for critical national infrastructure.
- Focused on multi-agent coordination and agentic reasoning frameworks.

Reviewer, LatinX in Computer Vision Workshop @ CVPR 2025

- Reviewed submissions focused on computer vision applications and methodologies.
- Emphasized promoting diversity and representation in AI research.

Reviewer, Second Workshop on Bangla Language Processing (BLP) @ IJCNLP-AACL 2025

- Reviewed submissions for the workshop co-located with IJCNLP-AACL 2025 in Mumbai, India.
- Contributed to advancing natural language processing research for Bangla language.

Reviewer, LatinX in AI (LXAI) Workshop @ NeurIPS 2025

- Evaluated papers and workshop submissions at prominent AI conference.
- Supported efforts to advance inclusion and emerging research within the AI community.

Ethics Reviewer, NeurIPS 2025 Datasets and Benchmarks Track

- Served as an ethics reviewer for dataset and benchmark papers.
- Ensured compliance with ethical research practices and responsible AI deployment.